

Adaptive Conformal Prediction for Full-Scale Plant pH Forecasting under Process Drift

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Abstract

Soft sensors in full-scale wastewater treatment face operating drift; overconfident point forecasts can mask compliance risk or force overly conservative chemical dosing. We evaluate conformal prediction intervals for 10-minute-ahead reactor pH forecasting on historical data from a semiconductor wastewater plant: a residual MLP trained on one source unit is evaluated offline across four anonymized units over multiple operating months.

Shift diagnostics relative to a source reference month show strong covariate separability (domain-classifier AUC) and large increases in relative MAE (RMAE), indicating model-error-scale drift under real operation. Comparing split, weighted, and online adaptive (ACI) conformal methods and an EnbPI-style rolling residual interval at nominal 90% coverage, we find static and covariate-weighted intervals undercover shifted regimes while online methods recover near-nominal coverage at moderate width cost.

Keywords: Conformal prediction, time-series forecasting, uncertainty quantification, covariate shift, industrial wastewater treatment

1. Setup and Shift Diagnostics

We use historical 5-minute data from four anonymized full-scale process units, exhibiting monthly operating changes, unit-to-unit differences, and process drift—a practical testbed for whether prediction intervals stay reliable after deployment.

The task is 10-minute-ahead Reactor 1 pH forecasting. A residual MLP, trained on Unit A and frozen, predicts the pH change from covariates available at prediction time—influent pH, influent flow, chemical dosing rate, and recent Reactor 1 pH lags—and is evaluated offline across all units and months.

Before applying conformal intervals, we diagnose deployment shift relative to Unit A July. Covariate shift is summarized by domain-classifier AUC, while prediction-error shift is summarized by RMAE, the unit-month MAE normalized by the Unit A July MAE. Figure 1 shows strong covariate separability and late-month error-scale inflation, especially in Units C–D. Because RMAE uses model errors, we treat it as a diagnostic for possible hidden or concept-like regime changes rather than as a pure covariate-shift metric.

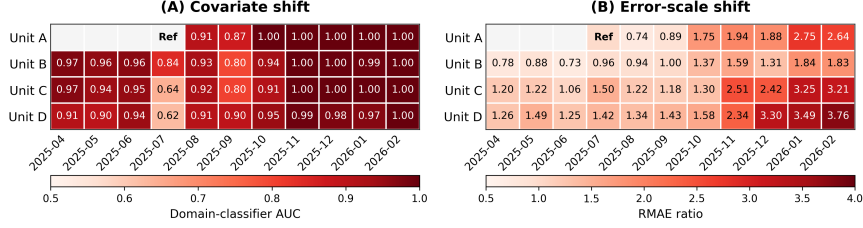


Figure 1: Shift diagnostics relative to the Unit A July reference. (A) Domain-classifier AUC quantifies covariate separability. (B) RMAE quantifies prediction-error scale relative to the reference block. Higher values indicate stronger covariate shift or larger error-scale inflation.

Method	Prot.	Cov.	Width	Score↓
Split CP	A	0.716	0.097	0.360
Split CP	B	0.747	0.110	0.334
Weighted CP	A	0.750	0.106	0.324
Weighted CP	B	0.827	0.132	0.300
ACI	B	0.898	0.165	0.251
EnbPI-style	B	0.890	0.147	0.255

Table 1: Aggregate UQ at nominal 90% coverage; lower interval score is better. ACI: $\gamma = 0.01$; EnbPI-style: β -optimized. Protocol A shown for static methods only.

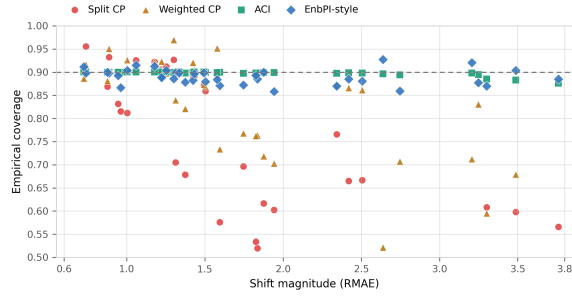


Figure 2: Per-unit-month coverage vs. shift (RMAE), Protocol B. Split CP and Weighted CP degrade with shift; online methods (ACI, EnbPI-style) stay near nominal 0.90 (dashed).

2. Conformal Interval Evaluation

We compare Split CP, Weighted CP under covariate shift (Tibshirani et al., 2019), ACI (Gibbs and Candès, 2021), and an EnbPI-style β -optimized rolling residual interval inspired by EnbPI (Xu and Xie, 2023). The EnbPI-style method uses a rolling residual window on a fixed MLP, not the original bootstrap ensemble.

Protocol A uses Unit A July as the calibration block for all units; Protocol B uses each target unit’s first available month. Online methods are reported under Protocol B. We report empirical coverage, mean interval width, and interval score, where lower values indicate sharper intervals with penalties for missed observations (Gneiting and Raftery, 2007).

Table 1 shows that static Split CP undercovered shifted regimes. Weighted CP improved coverage, especially with target first-month calibration, but remained below nominal. Online methods recovered near-nominal coverage. As RMAE grows, Split CP and Weighted CP degrade progressively, whereas ACI and EnbPI-style remain near the nominal 0.90 line (Figure 2). These results suggest that covariate reweighting helps but is insufficient under industrial residual drift; reliable long-term deployment requires online feedback from target observations rather than static source-reference intervals alone.

Acknowledgments

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